



DEVELOPMENT OF PALATE

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DIKIOHS

- 
- primary palate that is the caudal aspect of one of the processes involved in facial development – the frontonasal process
 - lies a common oronasal cavity that has to have secondary palate development before the oral and nasal cavities become separated.
 - secondary palate is derived from two medial extensions of the maxillary processes of the developing face – called palatal shelves

Palatal Shelves

- initially are orientated vertically at the sides of the developing tongue
- rapidly elevate into a horizontal position so that they can then fuse.
- palatal shelf elevation is caused primarily by hydration of the shelf mesenchyme, involving hyaluronan glycosaminoglycan.
- role of the mesenchymal cells??
- Involving the loss of a midline epithelial seam between the two fusing shelves.

By the 6th week of development

- The Frontonasal process forms primitive nasal cavities separated by a primary nasal septum and are partitioned from the primitive oral cavity by a primary palate.
- two lateral palatal shelves develop behind the primary palate from the maxillary processes
- secondary nasal septum grows down dividing the nasal part of the oronasal cavity into two

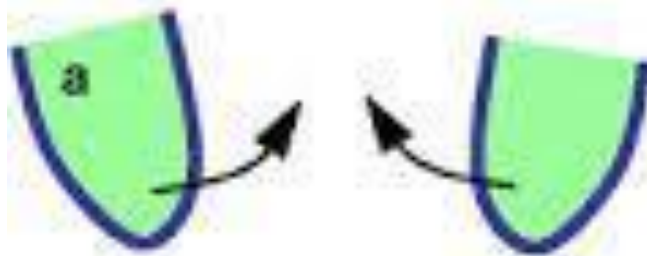
By 7th week

- the oral part of the oronasal cavity
- becomes completely filled by the developing tongue.
- palatal shelves continues to grow and lie vertically. (space in the oronasal cavity is insufficient for large tongue)
- Two peaks of DNA synthesis occur as the palatal shelves are formed: during initial shelf outgrowth and during vertical shelf elongation.

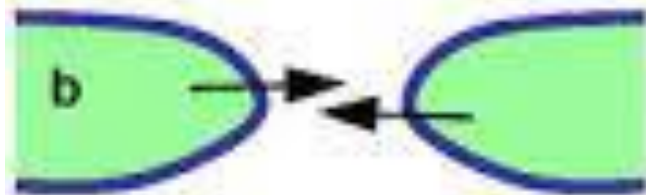


Fig. 18.3 Coronal section through the developing head during the 7th week of development showing the palatal shelves (A). B = developing tongue (Masson's trichrome; $\times 30$).

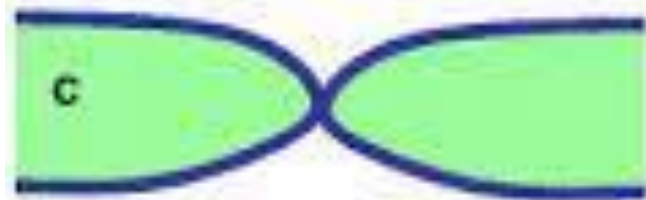
Main steps of palatogenesis:



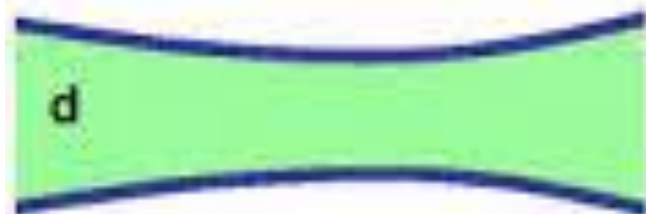
elevation



horizontal growth



midline contact,
adhesion



fusion
(confluence)

By 8th week

- stomodeum enlarges,
- the tongue 'drops' (*mandibular growth and/or a change in the shape of the tongue*)
- the vertically incline palatal shelves become horizontal
- the palatal shelves contact each other
- the medial edge epithelial seam degenerates so that mesenchymal continuity is established.
- Fusion of the palatal processes is complete by the 12th week of development.
- Behind the secondary nasal septum, the palatal shelves fuse to form the soft palate and uvula

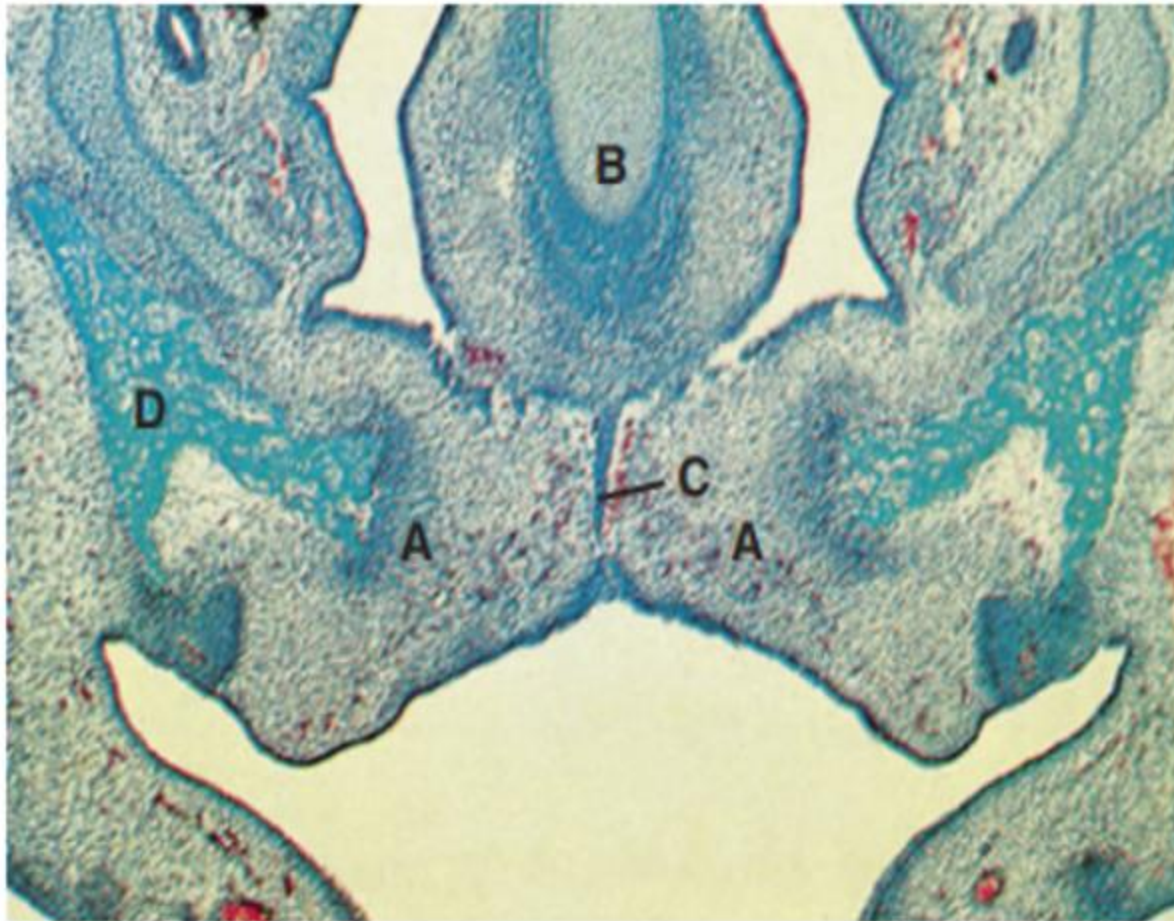


Fig. 18.5 Coronal section through developing oronasal regions following contact of the palatal shelves (A and secondary nasal septum (B). C = midline epithelial seam; D = developing bone of maxilla (Masson's trichrome; $\times 30$).

Palatal shelf elevation

- Extrinsic forces e.g., from forces derived from the tongue or jaw movements or as a result of differential pressures above and below the palatal shelves)

Intrinsic

- hydration of extracellular matrix components (principally hyaluronan) in the shelf mesenchyme
- Three GAG types are found in vivo: hyaluronan, heparan sulphate and chondroitin 4-sulphate
- It has been proposed that hyaluronan is a GAG involved in shelf elevation because it is a highly electrostatically charged, open-coil molecule capable of binding up to 10 times its own weight in water

Fusion of Palatal shelves

- Once the palatal shelves have elevated, they contact each other (initially in the middle third of the palate) and adhere by means of a 'sticky' glycoprotein, which coats the surface of the medial edge epithelia of the shelves.
- The epithelial cells develop desmosomes and consequently an epithelial seam is formed (Fig. 18.19).
- The adherence of the medial edge epithelia is specific as palatal epithelia will not fuse with epithelia from other sites (e.g., the tongue).
-

Ossification centres hard palate

- Once fusion is complete, the hard palate ossifies intramembranously
- from four centres of ossification
- , one in each developing maxilla and one in each developing palatine bone.
- The maxillary ossification centre lies above the developing deciduous canine tooth germ and appears in the 8th week of development (Fig. 18.5).
- The palatine centres of ossification are situated in the region forming the future perpendicular plate and appear in the 8th week of development.
- Incomplete ossification of the palate from these centres defines the median and transverse palatine sutures.
- Fig. 18.26 provides a coronal section through the developing hard palate
- to show early ossification.
- Areas that remain unossified in the hard palate provide bony canals for
- the passage of nerves and vessels, namely the **incisive canals** combining
- to form an **incisive fossa** between the primary and secondary palate in
- the **anterior midline**, and the **greater** and **lesser palatine canals** in the
- posterolateral region.
- During development, transient paired strands of
- epithelium are found in the region of the incisive canals termed **nasopalatine ducts**

- Palatal cysts-nasopalatine duct proliferation

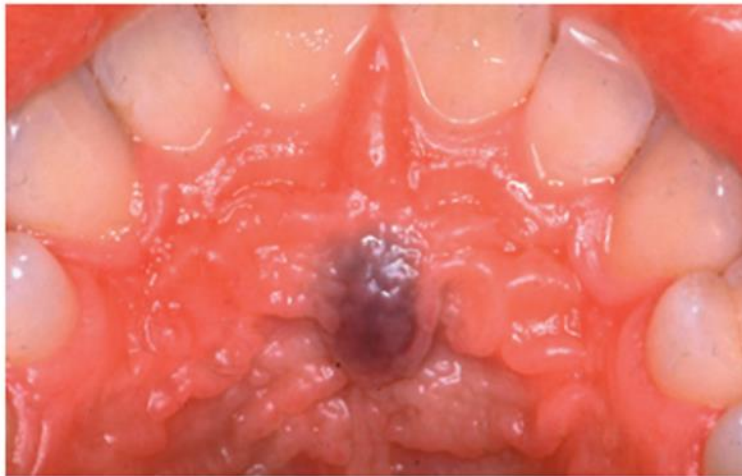


Fig. 18.27 Nasopalatine cyst evident as a bluish swelling in the anterior midline region of the palate. Courtesy Dr J Potts.



Fig. 18.28 Radiograph showing a heart-shaped radiolucent nasopalatine cyst in the anterior region of the maxillae. Courtesy Dr J Potts.

Palatal clefts

Table 18.1 Clefts of the upper lip and of the palate

Cleft lip	Cleft palate
Unilateral or bilateral	Classification with reference to incisive fossa – anterior anomalies (primary palate) and posterior anomalies (secondary palate)
With or without cleft palate	
1 : 1000 births	1 : 2500 births
Mostly male infants	More common in females
67% with multifactorial aetiology	41% with multifactorial aetiology
Result of failure of fusion of mesenchymes in facial processes	Result of failure of elevation of palatal shelves (major clefts) or failure of fusion of shelves (minor clefts)

submucous cleft

- where the palatal mucosa is intact but the bone/musculature of the palate is deficient beneath the mucosa

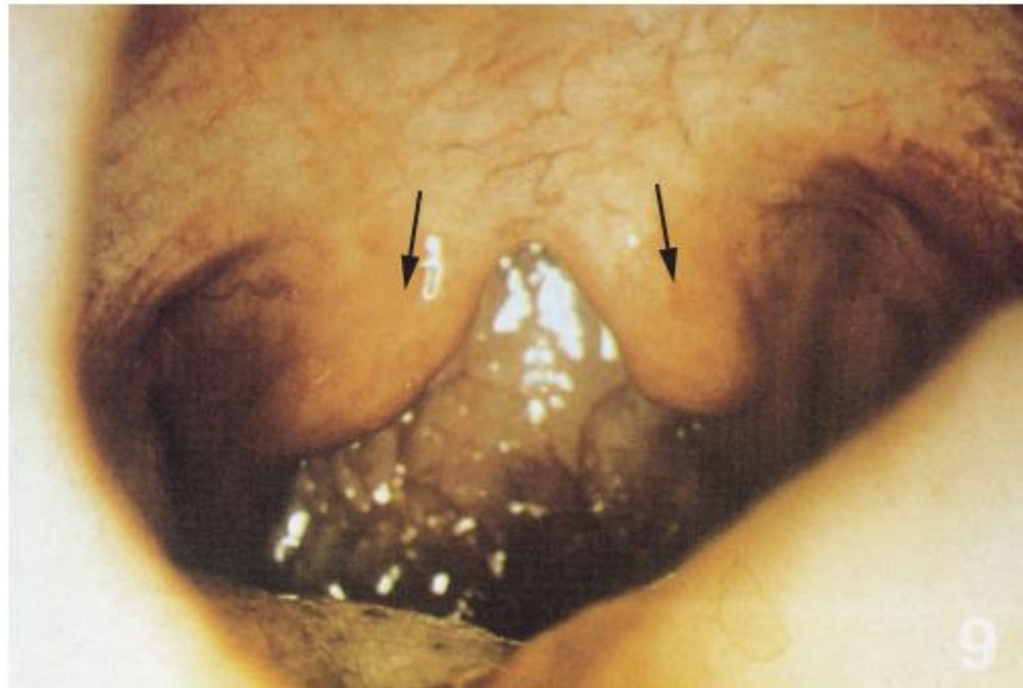


Fig. 18.30 A cleft uvula (arrows). Courtesy Dr R W Pigott.